

Learner-Dependent Cost of Mutual-Information Feature Subsets for Six-Class Intrusion Detection on CICIoMT2024

Siman Giri

Institute of Applied Sciences, Digital Technology Program

Madan Bhandari University of Science and Technology, Chitlang, Nepal

siman.giri@mbust.edu.np

Abstract—Internet of Medical Things traffic is multi-protocol and dominated by flood attacks. On the official CICIoMT2024 Wi-Fi/MQTT attack CSVs we ask whether four inspectable classifiers—logistic regression, k -nearest neighbours, a decision tree, and an RBF support vector machine—lose six-class macro- F_1 when 45 numeric flow statistics are replaced by the 15 features with highest mutual information with the label. The selector sits inside the training fold. The working matrix is a nested stratified sample of at most 2,000 flows per class (Benign, DDoS, DoS, MQTT, Recon, Spoofing), so that no flood shard can redefine DDoS as a single subtype. Exact duplicates and zero-variance columns are removed; median imputation and standardisation are fitted on training data only. Hold-out macro- F_1 on a 20% split is the decision metric, with a pre-registered equivalence band of 0.02. The decision tree is insensitive to the subset (macro- $F_1 = 0.948$ under both conditions, $\Delta = 0.000$). k -NN and the SVM lose 0.021 and 0.030. Logistic regression loses 0.139, driven by a fall in DoS F_1 from 0.637 to 0.357. Permutation overlap with the full-sample MI list tracks those gaps (11/15 for the tree, 7/15 for logistic regression). TreeSHAP on the winner concentrates on inter-arrival time, magnitude, and send rate—coordinates a univariate filter keeps—whereas the linear model was using joint geometry and protocol cues the filter under-ranks. A 15-feature MI subset is therefore sufficient for the tree on this sample and not sufficient for logistic regression. The claim does not extend to the official 19-class task, to Bluetooth traces, or to live hospital traffic.

Index Terms—CICIoMT2024, Internet of Medical Things, intrusion detection, mutual information, feature subset, decision tree, logistic regression, k -NN, SVM.

I. INTRODUCTION

The Internet of Medical Things (IoMT) is the set of networked devices that measure, actuate, or relay clinical signals: bedside monitors, infusion pumps, glucometers, wearables, and the lightweight brokers that publish their telemetry. Unlike general IoT sensors in a home or factory, these devices sit on a hospital or campus fabric, often with long service lives, vendor firmware, and constrained radios. Traffic is therefore mixed. Device telemetry commonly uses Wi-Fi for the last hop and MQTT for publish/subscribe messages; other radios (for example Bluetooth Low Energy) appear in the same wards but are not the subject of this study. A network observer sees *flows*—short windows of packets summarised by rates, flags, and protocol indicators—rather than a clinical record. Misclassifying those flows is an availability and integrity

problem for the device network: flood and reconnaissance traffic can starve or map a clinical segment without ever touching an electronic health record.

IoMT traffic of that kind is exactly what CICIoMT2024 records: 18 attacks collected from 40 medical and MQTT devices, released both as packet captures and as 45-dimensional flow tables [1], [2]. The authors already group the attacks into five families plus benign traffic. That six-class label—Benign, DDoS, DoS, MQTT, Recon, Spoofing—is the task of this paper.

Throughout, *Condition A* is the full numeric feature set after constant-column removal. *Condition B* is the 15-dimensional subset obtained by mutual information with the six-class label, with the selector fitted on each training fold only so that the hold-out set never chooses the features.

The rest of the paper is organised as follows. Section II reviews CIC flow corpora and prior use of CICIoMT2024. Section III describes the working sample. Section IV records the redundancy that motivates Condition B. Section V states the protocol. Section VI reports the 2×4 grid. Section VII reads DoS-DDoS leakage and the attribution overlap. Section VIII closes.

The contributions of this work are:

- A documented nested sample of the official CICIoMT2024 Wi-Fi/MQTT *attack* CSVs: at most 2,000 flows per six-class label, split across subtype shards so that DDoS cannot collapse to a single flood file.
- A leakage-safe comparison in which the *only* experimental factor is the feature set—all remaining numeric statistics (Condition A) versus the 15 features with highest mutual information with the label, selected on training folds only (Condition B)—under logistic regression, k -NN, a decision tree, and an RBF-SVM.
- An error analysis that treats residual DoS $\leftrightarrow$ DDoS confusion, after a per-class cap, as a property of windowed flow statistics rather than as a failed implementation.
- A post-hoc check that the MI-15 list does or does not coincide with permutation importance (all four models) and TreeSHAP (decision tree), so that a drop under Condition B can be attributed to discarded interactions rather than to chance.

We do not claim a new architecture, a result on the 19-class task, or a deployment on live hospital traffic.

II. RELATED WORK

Network intrusion research moved from packet dumps to CIC-style flow tables because a row of rates, flags, and protocol indicators can be classified with ordinary supervised learners. CICIDS2017 [3], UNSW-NB15 [5], and CICIoT2023 [4] are the immediate ancestors of CICIoMT2024. Surveys of that line already record the usual defects: label noise, near-duplicate rate features, and accuracy numbers that collapse once the majority flood class is down-weighted [6], [7]. Those defects are not historical. They reappear as soon as a medical-device corpus is released at CIC scale.

CICIoMT2024 is the first CIC table built on 40 medical and MQTT devices and 18 attacks spanning Wi-Fi, MQTT, and Bluetooth [1], [2]. The source paper already reports logistic regression, AdaBoost, random forest, and a small neural net on binary, six-class, and 19-class labels, and notes that 19-class recognition is the hard setting [1]. That is why this work stays at six classes: the authors' own family taxonomy, not a 19-way naming of flood subtypes.

Work that followed the release has mostly gone in two directions. The first is *more model capacity*. Doménech *et al.* show that a detector trained on CICIoT2023 loses a large fraction of F_1 when tested on CICIoMT2024, which is an argument for using the medical corpus rather than a generic IoT table [18]. Gheni and Al-Yaseen train logistic regression, AdaBoost, random forest, and several neural nets on CICIoMT2024 after an unspecified feature-selection step [19]. Hybrid deep models (BiLSTM-DNN and similar) report 19-class accuracies above 98% on the same files [20]. None of those papers asks whether the 45 flow statistics are necessary for the six-class task.

The second direction is *filter-based compression*. Rehman *et al.* compare Fisher score, mutual information, and information gain on CICIoMT2024 and IoMT-TrafficData, and find that a handful of features already suffice for binary detection, with about 15 features comfortable for multi-class detection under XGBoost [21]. That result is the closest prior to Condition B. It differs from the present experiment in three respects: the selector is not locked inside a training fold of a pre-registered 2×4 grid; the headline learners are ensembles rather than the four inspectable hypothesis classes used here; and there is no paired reading of permutation importance and TreeSHAP against the MI list. Mutual information itself is a standard univariate filter [8]; Shapley-style attributions [10] are used in this paper only to check that filter, not as a third feature-set condition.

The gap we take is therefore narrow. On the official Wi-Fi/MQTT *attack* CSVs, with a documented per-class sample, is a 15-feature MI subset enough for logistic regression, k -NN, a decision tree, and an RBF-SVM to hold six-class macro- F_1 within 0.02 of the full numeric vector? That question is not answered by a new deep encoder, and it is not answered by a filter study that never holds the rest of the protocol fixed.

III. DATASET AND PREPARATION

A. Source

We use the official CICIoMT2024 release [1], [2]. The download tree contains two experiments that must not be mixed. The `WiFi_and_MQTT/attacks` CSVs are labelled intrusion-detection traffic and are the population for this paper. The `profiling` captures (power, idle, active, interaction) record device lifecycle under benign use; they answer a fingerprinting question and are discarded. Bluetooth attack traces use a different schema and are discarded.

Only the authors' `train` shards are read. The official `test` split is left untouched so that a later external check remains possible; it is not used in the numbers below.

B. Files and labels

Each CSV is one attack subtype, or one chunk of a large flood. The six-class label is not stored as a column; it is inferred from the filename. Large floods are split across numbered files (`ICMP1`, `ICMP2`, ...). Those extra chunks are the same subtype, not new classes. We therefore keep one shard per flood subtype and every file of the smaller families.

The working download is:

- Benign: `Benign_train.pcap.csv`;
- Spoofing: `ARP_Spoofing_train.pcap.csv`;
- Recon: OS scan, ping sweep, port scan, vulnerability scan;
- MQTT: DDoS/DoS connect flood, DDoS/DoS publish flood, malformed data;
- DoS: one shard each of SYN, TCP, UDP, and ICMP;
- DDoS: one shard each of SYN, TCP, UDP, and ICMP.

An inventory of those 19 files recorded 45 columns and zero missing cells in every shard. Column counts were consistent across files.

The six-class map is: Benign; DDoS (distributed SYN/TCP/UDP/ICMP floods); DoS (the same four families from a single source); MQTT (connect and publish floods, malformed data); Recon (OS scan, ping sweep, port scan, vulnerability scan); Spoofing (ARP).

C. Feature families

CIC stores each flow as 45 numeric statistics extracted from a short packet window (10 or 100 packets, depending on the attack) [1]. We do not reprint that dictionary. For the experiments it is enough to treat the columns as four families.

Volume and timing. `Rate`, `Srate`, `Duration`, and inter-arrival time (`IAT`) describe how dense the window is. Flood classes live here; after a per-class cap they are less separable from one another than in the raw dump.

Size geometry. `Header_Length` and the magnitude / radius / variance moments summarise packet size. Several of these columns are nearly collinear, which is visible later as $|r| \geq 0.90$.

TCP flags. `syn_flag_number`, `ack_flag_number`, `fin_flag_number` and their siblings record handshake

TABLE I
WORKING-SAMPLE DESIGN.

Class	Official shards	Sampling rule	n
Benign	One train CSV	Cap from that file	2,000
Spoofing	ARP train CSV	Cap from that file	2,000
DDoS	Four flood subtypes	Equal quota, then top-up	2,000
DoS	Four flood subtypes	Equal quota, then top-up	2,000
MQTT	Five MQTT files	Equal quota, then top-up	2,000
Recon	Four recon files	Equal quota; all if short	$\leq 2k$

structure. A SYN flood is almost a one-hot flag vector; that is why a tree can look strong without using rates.

Protocol indicators. Protocol Type and the binary protocol columns (TCP, UDP, ICMP, MQTT, HTTP, ...) mark which stack produced the window. They are the cheapest six-class cues and the reason 19-class flood naming is easy.

Condition A uses every numeric column that survives the constant-feature audit. Condition B keeps 15 coordinates chosen by mutual information with the six-class label. The families above explain why 15 can be enough: the filter is largely choosing one representative from each redundant group, not 15 unrelated sensors.

D. Nested stratified sample

The raw shards are too large, and too skewed toward floods, for a four-model grid in a notebook. We therefore build an analysis sample of at most 2,000 flows per six-class label, seed 42.

Sampling has two levels. The first level is the target class, so that Benign, DDoS, DoS, MQTT, Recon, and Spoofing are equally represented by design. The second level is the official shard inside that class. Benign and Spoofing have one file; the class budget is drawn from that file. Multi-file classes receive an equal per-file quota ($\lfloor 2000/n_{\text{files}} \rfloor$), clipped to the file length. Any unused quota is filled from leftover rows of the larger shards. Ping sweep, with 740 rows in the inspected train file, contributes every clean row and may leave Recon slightly under 2,000.

This rule is the difference between a balanced six-class study and an accidental ICMP-only “DDoS” class. The six class samples are concatenated, shuffled with the same seed, and written to disk. All later EDA and models read only that file.

E. Cleaning on the working table

Non-numeric fields are coerced; $\pm\infty$ is replaced with NaN. Exact duplicate rows are removed because they overweight a single window in k -NN and in accuracy. Columns with one unique value are removed because they have zero variance and break standardisation. Near-constant binary flags are kept: they are legitimate protocol indicators. No synthetic minority rows are inserted. No test-set information is used in these decisions. Figure 1 records the class counts on that cleaned table. The

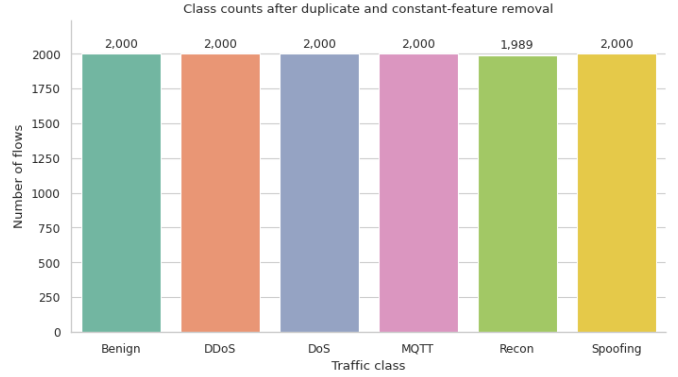


Fig. 1. Class distribution after cleaning. Each bar is one of the six target labels. Height is the number of flows retained after exact-duplicate removal and deletion of constant features. Bars sit near 2,000 by design; a short Recon bar means ping sweep could not fill its quota, not that the class was dropped.

cleaned table—about 12,000 rows, six labels, the surviving numeric features plus label and source_file—is the only matrix used from Section IV onward. Replace “about 12,000” with the integer printed after your cleaning cell.

IV. EXPLORATORY ANALYSIS

All analysis in this section uses only the cleaned working table. Class counts after de-duplication appear in Fig. 1; they are nearly uniform by construction. A short Recon bar is a depleted shard (ping sweep), not a modelling choice. A stacked composition of official shards (the sampling log) confirms that DDoS, DoS, MQTT, and Recon remain mixtures rather than single-file artefacts.

A. Scale and redundancy

Volume and timing columns live on incompatible raw scales; many flag and protocol columns are almost binary. That is why every later pipeline contains a scaler, including the decision tree: the comparative factor is the feature set, not an inconsistent preprocessor.

Pearson correlation on the numeric matrix yields many pairs with $|r| \geq 0.90$, concentrated inside the rate family and inside the size-moment family (Fig. 2). Those blocks are the *a priori* case for Condition B: a univariate filter can drop a near-duplicate without discarding a new signal. We do not delete correlated columns by hand. The subset experiment does that under a fixed rule.

B. Descriptive mutual information

Mutual information with the six-class label is estimated on the full working sample with a k -nearest-neighbour estimator [8]. Protocol indicators, selected TCP flags, and a small number of rate / inter-arrival columns sit at the top of Fig. 3; near-zero columns are weak family cues.

This list is not Condition B. Condition B re-estimates the same score on each training fold so that the hold-out set never chooses the 15 features. Agreement between the two lists is checked after the models are fit, not here.

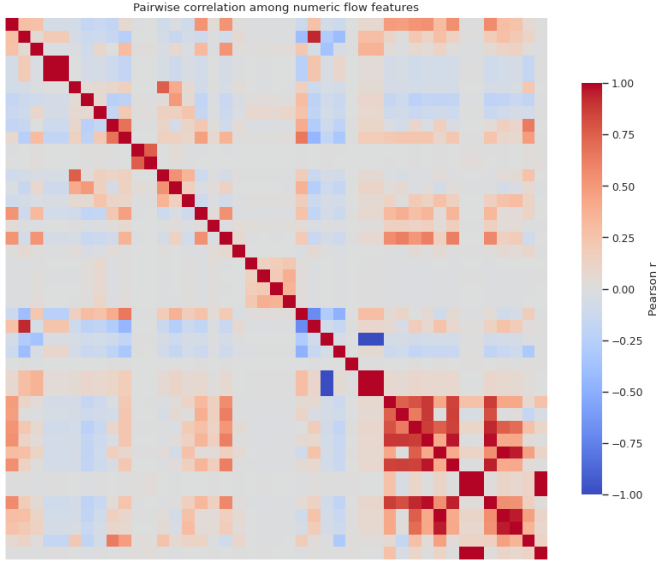


Fig. 2. Pairwise Pearson correlation among numeric flow statistics on the cleaned working sample. Axis labels are omitted; 45 names are unreadable at column width. Dark off-diagonal blocks are the near-duplicate rate and size-moment families that Condition B is allowed to compress.

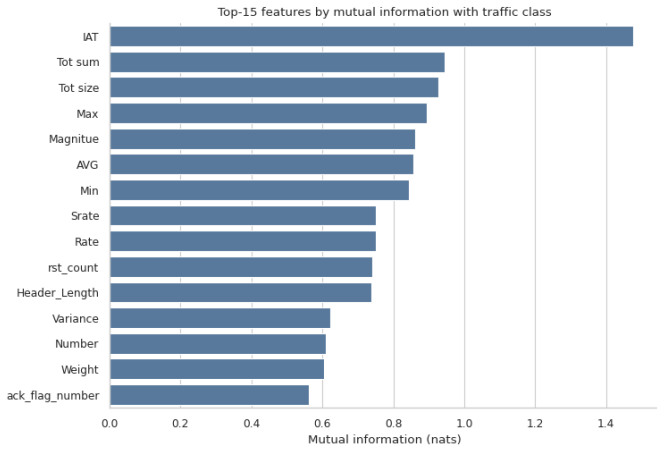


Fig. 3. Top-15 flow statistics by mutual information with the six-class label, estimated on the full cleaned working sample. Higher bars share more information with {Benign, DDoS, DoS, MQTT, Recon, Spoofing}. The experimental subset is re-selected inside each training fold; this figure is descriptive only.

C. Geometry

A two-dimensional PCA of the standardised matrix [12] is shown in Fig. 4. Class structure is visible, with overlap between DoS and DDoS. The first two components explain only a fraction of scaled variance; the scatter is a view, not a proof of separability. A stratified t -SNE snapshot [11] of 4,000 flows tightens local neighbourhoods but does not preserve global distances. Neither embedding is used as input to logistic regression, k -NN, the decision tree, or the SVM.

Class-conditional boxplots of a few high-MI columns (outliers hidden, separate y -scales) are a sanity check in the notebook: SYN-flag mass rises in SYN floods; protocol type

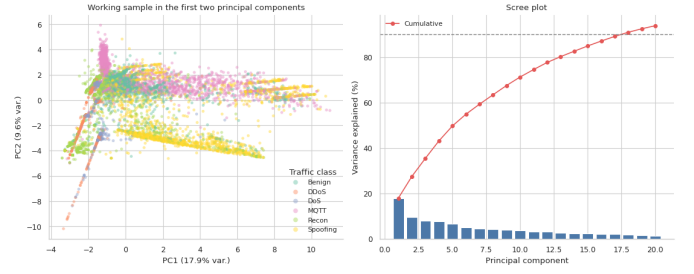


Fig. 4. First two principal components of the standardised working sample. Colour is the six-class label. DoS and DDoS overlap; that geometry reappears in the confusion matrices and is treated as a property of windowed flow statistics, not as a broken pipeline.

splits ICMP/UDP/TCP families. They are omitted here to keep the section to three figures.

V. EXPERIMENTAL SETUP

A. Matrix and split

Let X be the numeric matrix after constant-feature removal and y the six-class label {Benign, DDoS, DoS, MQTT, Recon, Spoofing}. A stratified 80/20 split with seed 42 produces a training block and a hold-out test set. The test set is not used for imputation statistics, scaling, feature selection, or hyper-parameter choice.

On the training block, stratified five-fold cross-validation maximises macro- F_1 . The winning specification is then refit on the full training block and scored once on the hold-out set.

B. Pipeline

Every experiment is a scikit-learn pipeline [13] with a fixed prefix: median imputation, standard scaling, optional SelectKBest, then the classifier. Scaling is applied to all four models, including the decision tree, so that the only changing factor is the feature set.

- *Condition A* omits the selector and uses every remaining numeric column.
- *Condition B* inserts `SelectKBest(mutual_info_classif, k = 15)` [8] before the classifier. The selector is fitted on the training portion of each fold. The hold-out set never chooses the subset.

The eight cells of the study are therefore four learners $\times$ two feature conditions.

C. Learners and grids

The four hypothesis classes are logistic regression [14], k -nearest neighbours [15], an axis-aligned decision tree [16], and an RBF support vector machine [17]. Grids are small by design (Table II). They exist to stop a single default hyper-parameter from being mistaken for a feature-set effect. They are not a claim that the global optimum was found.

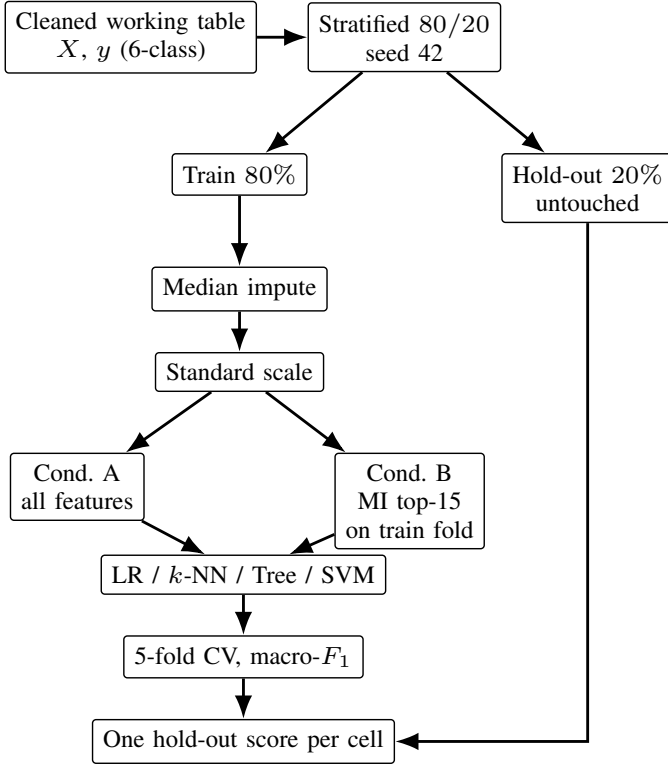


Fig. 5. Experimental protocol. Imputation, scaling, and the Condition B selector are fitted on training folds only. Four learners $\times$ two feature conditions give the eight cells in Section VI.

TABLE II
LEARNERS AND HYPER-PARAMETER GRIDS.

Learner	Hypothesis class	Grid
Logistic regression	Linear, multinomial	$C \in \{0.1, 1, 10\}$
k -NN	Local distance	$k \in \{3, 5, 11\}$
Decision tree	Axis-aligned splits	depth $\{5, 10, \text{None}\}$; leaf $\{1, 5\}$
RBF-SVM	Maximum-margin kernel	$C \in \{1, 10\}$; $\gamma \in \{\text{scale}, 0.01\}$

D. Metrics and decision rule

The primary score is macro- F_1 . Accuracy and weighted- F_1 are reported but do not decide the comparison: the working sample is balanced by construction, the original CIC prior is not. Per-class F_1 and the hold-out confusion matrix are reserved for the error analysis.

We pre-register a practical equivalence band of 0.02 on hold-out macro- F_1 . Condition B is treated as competitive with Condition A for a given learner when $|\Delta| = |F_1^B - F_1^A| < 0.02$ on the same split. Class weights are left off. The per-class cap already removes the flood prior; re-weighting would answer a different question.

Permutation importance and TreeSHAP are computed after the eight cells are fit. They are not part of model selection.

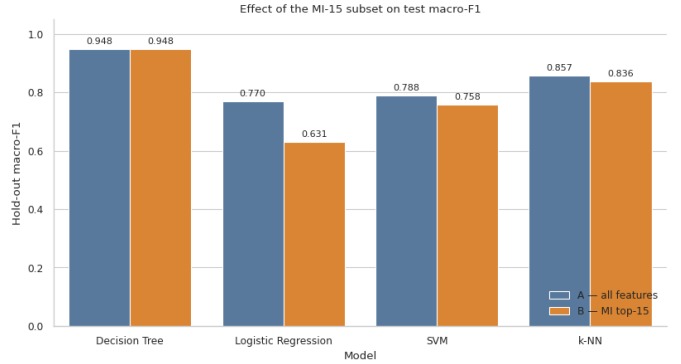


Fig. 6. Hold-out macro- F_1 for the eight cells. Each pair is one learner; the right-hand bar is Condition B. The tree pair is flat; the logistic-regression pair is not.

VI. RESULTS

We report four measurements, in the order they answer the experimental question. All scores use the same hold-out split. Macro- F_1 is the decision metric.

A. Full-feature baselines

On Condition A the four learners separate cleanly (Table III, rows marked A). The decision tree reaches a hold-out macro- F_1 of 0.948, k -NN 0.857, the RBF-SVM 0.788, and logistic regression 0.770. Cross-validation tracks the hold-out scores to two decimals, so the ranking is not a split artefact. The tree is therefore the baseline against which the subset must be judged: if Condition B is to be called sufficient, it has to keep this model near 0.948, not merely keep logistic regression from collapsing.

B. Cost of the MI-15 subset

Condition B repeats the same four learners with the selector inside the pipe (Table III, rows marked B; Fig. 6). The signed cost $\Delta = F_1^B - F_1^A$ is not uniform.

The tree is insensitive: $\Delta = 0.000$. k -NN loses 0.021 and sits on the pre-registered 0.02 band. The SVM loses 0.030. Logistic regression loses 0.139 (macro- F_1 falls from 0.770 to 0.631). Model ranking does not change—tree, k -NN, SVM, logistic regression—but the linear model is no longer a serious six-class detector once the features are filtered univariately.

The practical sentence is therefore conditional. A 15-feature MI subset is interchangeable with the full numeric vector for the decision tree on this sample. It is not interchangeable for logistic regression. Distance and kernel models sit in between.

C. Selected hyper-parameters

Table IV records what five-fold CV actually chose. SVM and logistic regression keep the same regularisation under both conditions ($C = 10$; $\gamma = \text{scale}$), so their Δ cannot be blamed on a different C . k -NN moves from $k = 5$ to $k = 11$ under the subset, which is consistent with a thinner neighbourhood after coordinates disappear. The tree is unrestricted in depth under Condition A and depth-10 under Condition B; hold-out F_1 does not move.

TABLE III
EIGHT-CELL GRID ON THE CLEANED WORKING SAMPLE. Δ IS THE CHANGE IN *test* MACRO- F_1 FROM CONDITION A TO CONDITION B.

Model	Cond.	CV macro- F_1	Test acc.	Test macro- F_1	Test w- F_1	Δ (B-A)
Decision tree	A	0.946	0.948	0.948	0.948	—
Decision tree	B	0.944	0.947	0.948	0.948	0.000
k -NN	A	0.847	0.857	0.857	0.857	—
k -NN	B	0.828	0.835	0.836	0.836	-0.021
SVM	A	0.788	0.788	0.788	0.788	—
SVM	B	0.756	0.766	0.758	0.758	-0.030
Logistic regression	A	0.763	0.769	0.770	0.770	—
Logistic regression	B	0.611	0.642	0.631	0.631	-0.139

TABLE IV
HYPER-PARAMETERS SELECTED BY FIVE-FOLD CV ON THE TRAINING BLOCK.

Model	Cond. A	Cond. B
Decision tree	unrestricted depth	depth 10
k -NN	$k = 5$	$k = 11$
SVM	$C = 10, \gamma = \text{scale}$	$C = 10, \gamma = \text{scale}$
Logistic regression	$C = 10$	$C = 10$

TABLE V
WEAKEST HOLD-OUT CLASS OF EACH CELL (CLASS-WISE F_1).

Model	Cond.	Weakest class	F_1
Decision tree	A	Spoofing	0.892
Decision tree	B	Spoofing	0.875
k -NN	A	Spoofing	0.775
k -NN	B	Spoofing	0.761
SVM	A	DoS	0.586
SVM	B	DoS	0.478
Logistic regression	A	DoS	0.637
Logistic regression	B	DoS	0.357

D. Where the remaining error sits

Macro- F_1 averages six labels. Table V names the worst label of each cell; Fig. 7 shows the other five.

Two families of failure appear. The tree and k -NN are limited by Spoofing under both conditions (0.892 and 0.775 on Condition A; 0.875 and 0.761 on Condition B). ARP is the smallest official family in the download; a 15-feature filter does not create that weakness and does not much worsen it.

SVM and logistic regression are limited by DoS under both conditions. That is the flood twin already visible in PCA. Condition B then treats the two learners differently. SVM’s weakest-class F_1 falls from 0.586 to 0.478. Logistic regression’s DoS F_1 falls from 0.637 to 0.357. The headline $\Delta = -0.139$ for logistic regression is therefore not a uniform trim across six labels; it is largely a collapse of the DoS column after univariate MI has removed coordinates the linear model was using to separate single-source floods from distributed ones.

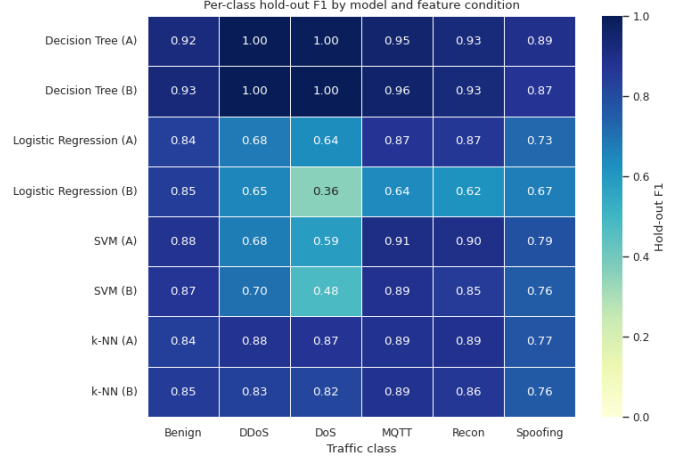


Fig. 7. Per-class hold-out F_1 . Each row is one cell of the grid. A pale DoS or DDoS column is source-multiplicity error, not a missing MQTT detector.

E. Permutation importance versus the MI list

The MI-15 list is a univariate ranking of X against y . It need not be what a fitted classifier uses. Table VI compares each Condition A model’s top-15 permutation features with that list. Fig. 8 shows the underlying ranks.

Overlap tracks Δ . The tree shares 11 of 15 features with the full-sample MI list and loses no hold-out macro- F_1 . k -NN and SVM share 8 of 15 and lose 0.021 and 0.030. Logistic regression shares 7 of 15 and loses 0.139. The extra permutation features for logistic regression—Covariance, Radius, Protocol Type, TCP—are interacting geometry and protocol cues that a univariate filter can under-rank. They are also the cues one would use to tell a single-source flood from a distributed one.

F. TreeSHAP on the winning model

Permutation importance is model-agnostic and global. For the decision tree we also compute mean absolute SHAP values on the hold-out set [10], so that the winner’s splits can be read without mixing them with logistic regression.

Table VII and Fig. 9 show a short head, not a flat list. IAT alone carries 0.146; magnitude (Magnitude in the CIC export) carries 0.096; Srate carries 0.040. After Header_Length and

TABLE VI
OVERLAP OF EACH MODEL’S TOP-15 PERMUTATION FEATURES
(CONDITION A) WITH THE FULL-SAMPLE MI-15 LIST.

Model	Overlap	Permutation-only features
Decision tree	11/15	TCP, fin_count , psh_flag_number , syn_count
k -NN	8/15	ICMP, Protocol Type, TCP, UDP, rst_flag_number
SVM	8/15	Duration, HTTP , ICMP, fin_count , rst_flag_number
Logistic regression	7/15	Covariance, Protocol Type, Ra- dius, TCP, fin_count

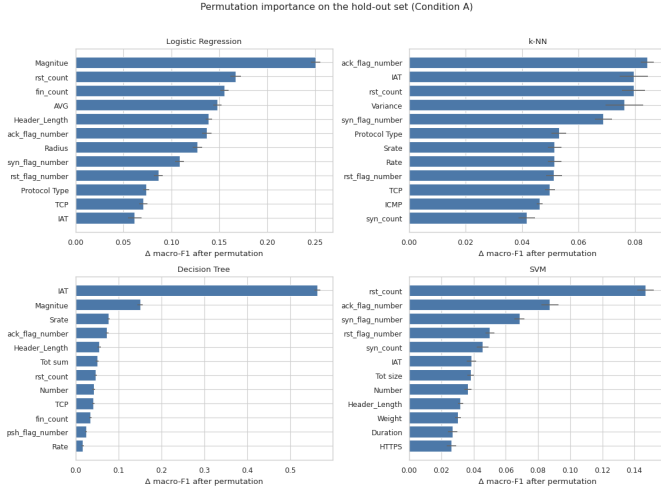


Fig. 8. Hold-out permutation importance under Condition A (drop in macro- F_1 when a column is shuffled).

rst_count the mass falls off. The tree is therefore a volume-and-timing model with a few flag counters, not a 45-way vote. Condition B can keep that head and still match Condition A ($\Delta = 0.000$). The same head is not what logistic regression was using to separate DoS from DDoS, which is why the two Δ values disagree.

KernelSHAP on SVM and k -NN was not run.

VII. DISCUSSION

A. What the grid decided

The pre-registered question was whether a 15-feature MI subset preserves six-class macro- F_1 within 0.02 of the full numeric vector. The answer depends on the hypothesis class.

For the decision tree the subset is interchangeable ($\Delta = 0.000$, hold-out macro- $F_1 = 0.948$ under both conditions). For k -NN the loss sits on the band (-0.021). For the RBF-SVM it is just outside (-0.030). For logistic regression the subset is not a compressor; it is a different, worse detector (-0.139). Ranking does not change. A reader who reports only the tree, or only accuracy, would have concluded that “15 features suffice.” That sentence is true for one of four models on this sample and false for another.

TABLE VII
TREESHAP MEAN |VALUE| FOR THE CONDITION A DECISION TREE ON
THE HOLD-OUT SET. MAGNITUDE IS THE CIC COLUMN NAME.

Rank	Feature	Mean SHAP
1	IAT	0.146
2	Magnitude	0.096
3	Srate	0.040
4	Header_Length	0.027
5	rst_count	0.026
6	ack_flag_number	0.016
7	Number	0.016
8	Tot sum	0.015
9	TCP	0.013
10	Variance	0.006

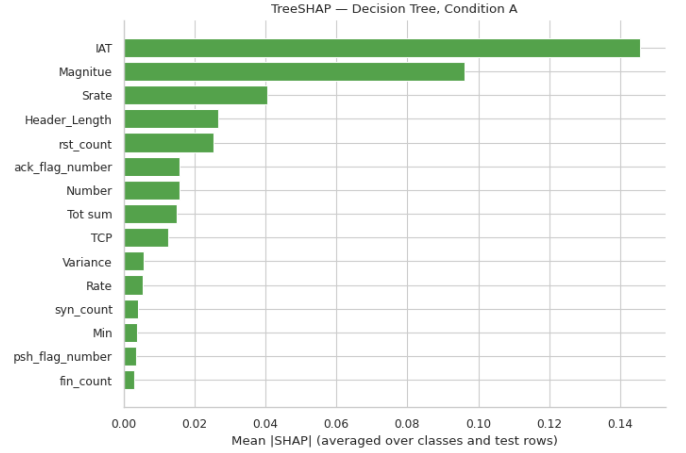


Fig. 9. TreeSHAP mean |value| on the Condition A decision tree. IAT, magnitude, and *Srate* dominate; the tail is small.

B. Why the tree survives and the linear model does not

Permutation overlap with the full-sample MI list tracks those four Δ values: 11/15 for the tree, 8/15 for k -NN and the SVM, 7/15 for logistic regression (Table VI). TreeSHAP on the winner is concentrated on IAT, magnitude, and *Srate* (Table VII). Those are volume-and-timing coordinates. A univariate filter that keeps the MI head keeps the tree’s splits, so Condition B looks like Condition A.

Logistic regression’s extra permutation features—Covariance, Radius, Protocol Type, TCP—are joint geometry and protocol cues. Univariate MI can under-rank them because each column, taken alone, is only a moderate class signal. The linear model was using the combination, in particular to separate DoS from DDoS. When the selector drops that combination, DoS F_1 falls from 0.637 to 0.357 and the headline Δ follows. That is a limitation of a univariate filter, not evidence that the eight pipelines were fit incorrectly.

Rehman *et al.* already observed that a small filter subset can look sufficient on CICIoMT2024 under ensembles [21]. The present grid says the same sentence must be qualified by hypothesis class: an ensemble or a deep tree can survive MI-15; a linear model on the same sample may not.

The weakest-class table is two problems, not one (Table V). The tree and k -NN are limited by Spoofing under both conditions. ARP is the smallest official family in the download; the subset does not create that floor and does not much move it ($0.892 \rightarrow 0.875$ for the tree; $0.775 \rightarrow 0.761$ for k -NN).

SVM and logistic regression are limited by DoS under both conditions. CIC flow statistics describe a short window. A single-source SYN flood and a distributed SYN flood share protocol number, flag profile, and size geometry. In the raw dump they still differ by volume. The per-class cap, which is required for a balanced six-class study, removes that cue (Fig. 4). Residual DoS $\leftrightarrow$ DDoS leakage is therefore a property of the representation after sampling, and it is the ceiling on macro- F_1 for any linear or kernel model that has to live on these 45 columns. Confusion matrices in the notebook show that pattern; they are not reprinted here.

D. Limitations

The analysis sample is not the official CIC dump: at most 2,000 flows per class, train shards only, one flood file per subtype. A different cap, or the official test split as an external check, could move the Δ values. The MI estimator is univariate; a wrapper on the training folds only might have kept Covariance and Radius for logistic regression and closed part of the -0.139 gap. Hyper-parameter grids are small. The 19-class task is out of scope. Nothing here is a hospital deployment, a Bluetooth result, or a claim that 15 features suffice for every IoMT detector. s do not transfer, without a new sample, to Bluetooth-only IoMT or to a live hospital trace.

VIII. CONCLUSION

This paper asked whether a 15-feature mutual-information subset preserves six-class macro- F_1 on a documented CIC-IoMT2024 Wi-Fi/MQTT working sample, for four inspectable learners, within a pre-registered band of 0.02. The subset is interchangeable with the full numeric vector for the decision tree ($\Delta = 0.000$, hold-out macro- $F_1 = 0.948$) and is not interchangeable for logistic regression ($\Delta = -0.139$). k -NN and the RBF-SVM sit between those poles (-0.021 , -0.030); model ranking does not change. The tree survives because its splits concentrate on IAT, magnitude, and S_{rate} , which univariate MI keeps; the linear model fails because it was using joint geometry and protocol cues that the filter under-ranks, and DoS F_1 then falls from 0.637 to 0.357. Two error floors remain under both conditions: Spoofing for the tree and k -NN, DoS for SVM and logistic regression—the second of these is expected once a per-class cap removes the volume cue that used to separate single-source floods from distributed ones. The result is tied to a 12k nested sample of official train shards and to a univariate selector; a training-fold wrapper, or the official test split as an external check, is the direct next measurement, not a new architecture.

Notebook order: setup; inventory of official CSVs; nested sample; EDA and cleaning; experimental objects; eight GridSearchCV fits; A-versus-B figure; per-class F_1 ; confusion matrices; Condition-B support masks; permutation importance; TreeSHAP on the tree. Artefacts live under results/: file_inventory.csv, sampling_log.csv, ciciomt2024_6class_clean.csv, model_comparison.csv, delta_macro_fl.csv, per_class_fl.csv, dos_ddos_leakage.csv, condition_B_selected_features.csv, permutation_importance_A.csv, treeshap_decision_tree_A.csv. Seed 42 throughout. Software: Python 3, pandas, scikit-learn, matplotlib/seaborn, shap (tree only). A large language model was used to check English and to help organise section wording; every experimental number is taken from the notebook artefacts above, and the authors remain responsible for the protocol, the fits, and the claims.

REFERENCES

- [1] S. Dadkhah, X. Neto, R. Ferreira, R. C. Molokwu, S. Sadeghi, and A. A. Ghorbani, "CICIoMT2024: A benchmark dataset for multi-protocol security assessment in IoMT," *Internet of Things*, vol. 28, Art. no. 101351, Dec. 2024.
- [2] Canadian Institute for Cybersecurity, "IoMT dataset 2024," Univ. New Brunswick. [Online]. Available: <https://www.unb.ca/cic/datasets/iomt-dataset-2024.html>
- [3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. ICISSP*, Funchal, Portugal, 2018, pp. 108–116.
- [4] E. C. P. Neto *et al.*, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, Art. no. 5941, 2023.
- [5] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. MilCIS*, Canberra, Australia, 2015, pp. 1–6.
- [6] M. Ring, S. Wunderlich, D. Scheuring, D. Landes, and A. Hotho, "A survey of network-based intrusion detection data sets," *Comput. Secur.*, vol. 80, pp. 147–167, Jan. 2019.
- [7] A. Khraisat, I. Gondal, P. Vamplew, and J. Kamruzzaman, "Survey of intrusion detection systems: Techniques, datasets and challenges," *Cybersecurity*, vol. 2, Art. no. 20, 2019.
- [8] A. Kraskov, H. Stögbauer, and P. Grassberger, "Estimating mutual information," *Phys. Rev. E*, vol. 69, no. 6, Art. no. 066138, 2004.
- [9] L. S. Shapley, "A value for n -person games," in *Contributions to the Theory of Games II*. Princeton, NJ, USA: Princeton Univ. Press, 1953, pp. 307–317.
- [10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.
- [11] L. van der Maaten and G. Hinton, "Visualizing data using t -SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.
- [12] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Philos. Trans. Roy. Soc. A*, vol. 374, Art. no. 20150202, 2016.
- [13] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [14] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.
- [15] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. 13, no. 1, pp. 21–27, Jan. 1967.
- [16] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Belmont, CA, USA: Wadsworth, 1984.
- [17] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.

- [18] J. Doménech, O. León, M. S. Siddiqui, and J. Pegueroles, "Evaluating and enhancing intrusion detection systems in IoMT: The importance of domain-specific datasets," *Internet of Things*, vol. 32, Art. no. 101631, Jul. 2025.
- [19] H. Q. Gheni and W. L. Al-Yaseen, "Using CICIOMT2024 dataset for improved intrusion detection system," in *Proc. ICDAM, 2024, LNNS*, vol. 1302. Singapore: Springer, 2025, pp. 285–301.
- [20] "HBiLD-IDS: An efficient hybrid BiLSTM-DNN model for real-time intrusion detection in IoMT networks," *Information*, vol. 16, no. 8, Art. no. 669, Aug. 2025.
- [21] M. U. Rehman, R. Kalakoti, and H. Bahşi, "Comprehensive feature selection for machine learning-based intrusion detection in healthcare IoMT networks," in *Proc. ICISSP, Porto, Portugal, 2025*, pp. 248–259.